

Editable Body : Interactive Adaptation of Avatar Control Schemes and Body Structures

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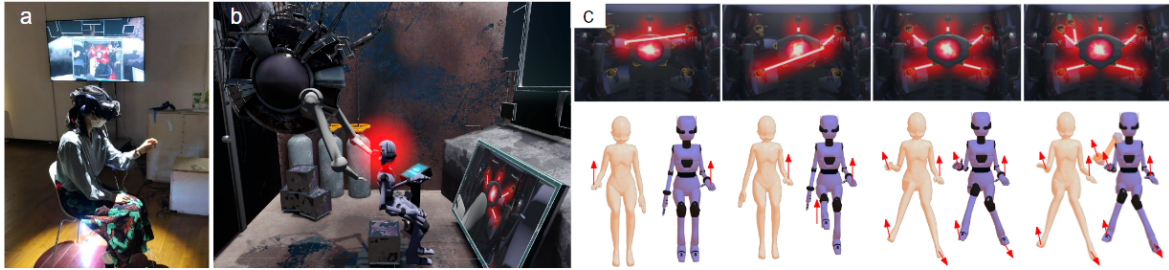


Figure 1: a) A player engaged in *Mechanical Brain Hacking*, modifying their avatar's body control and structure in real time. b) A game scene where the player manipulates the avatar's brain using a surgical machine. c) The player's limb-to-avatar mapping is set by brain circuits. The player can also add a third arm to the avatar, as shown on the rightmost side.

ABSTRACT

In VR, users can switch between avatars with different structures and control schemes. Previous research on body perception has shown that users quickly adapt to controlling new bodies, a process known as body schema learning. However, there is a lack of studies and games that support real-time, user-driven body modifications. To address this gap, we present *Mechanical Brain Hacking*, a VR game based on the concept of the "Editable Body," which enables users to modify the structure and control scheme of their avatars in real time. We investigated the user experience of this interactive body modification system and explored its effect on users' perceptions of their own bodies and attitudes toward body modification technologies. Our findings indicate that this system leads to greater acceptance of identity expression through non-innate bodies, encourages users to view the body as more fluid and adaptable, and increases openness to body modification technologies.

Index Terms: Non-humanoid Avatar, Embodiment, Body Schema, Virtual Reality, Adaptive Embodiment

1 INTRODUCTION

In virtual reality (VR), users can wear non-human modeled avatars—avatars whose appearance and structure differ from humans—to create novel, captivating experiences. These include avatars that differ only in appearance (e.g., gorillas) and those that diverge in structure (e.g., birds, octopuses) [1]. In social VR such as VRChat, many users create and wear avatars with different levels of anthropomorphised avatars [4], suggesting the allure and potential of non-human embodied experiences as a new VR genre [2].

While VR users can easily switch avatars, few systems or studies allow for interactive customization of an avatar's body structure and control scheme. Previous studies have shown that people can embody different body structures and manipulate them in various ways, as illustrated in Fig. 2. In particular, research on hu-

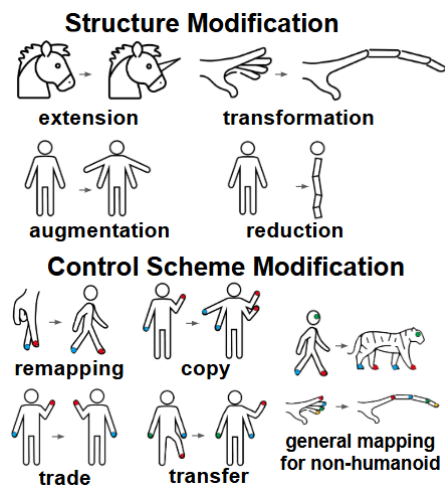


Figure 2: Overview of Common Avatar Body Structures and Control Scheme Modifications

mans' ability to adapt to novel bodies has demonstrated that they can quickly adjust to new body configurations [5], indicating the potential for short-term plasticity in virtual bodies. If users had an "editable body" — a body that allows real-time modification of the avatar's structure and control scheme as putting on or removing accessories — it would not only offer novel gameplay experiences but also open new perspectives in researches about embodiment.

Our novel gaming system, *Mechanical Brain Hacking* (MBH) (Fig. 1), addresses this gap by enabling users to interactively modify both the control scheme and structure of their VR avatars. Users adopt a robotic avatar with "broken" control circuitry, which they must repair and extend by adding a third arm. A telepresence-style surgical arm allows direct manipulation of the avatar's brain circuits, which link the user's real limbs to the avatar's limbs (Fig. 1c). By rewiring these circuits, users restore normal limb mappings and may also attach new components, reassigning their real limbs to control the newly added parts.

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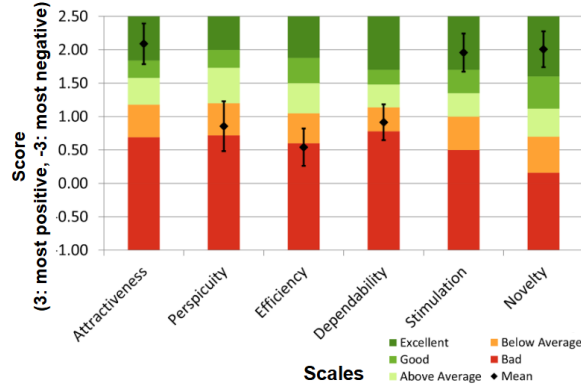


Figure 3: Results of the UEQ for each scale, with benchmark evaluations. The error bars indicate the 95% confidence interval.

2 SYSTEM & SCENARIO

In MBH, the participant controls a robot avatar whose internal circuits defines limb-to-avatar mapping. A surgical arm, moving in tandem with the robot's arm, lets participants rewire the avatar's brain while viewing a display of the avatar's head (Fig. 1b). In the control circuit, each user limb corresponds to an input terminal (e.g., the user's right arm), and each corresponding avatar limb corresponds to an output terminal (e.g., the avatar's right arm). By connecting or disconnecting beams of light between terminals, users immediately change the mappings. Initially, the connections are misaligned (e.g., the right arm controls the avatar's left arm). The user's first goal is to restore normal mapping; once achieved, the user can attach a third arm to the avatar and map any of their limbs to control it. Throughout the process, a vibrotactile stimulus on the back of the user's head indicates when the surgical arm makes contact, and an electrical stimulus is applied to the relevant limbs during rewiring. Participants found a comfortable mapping to operate the third arm and completed the roughly 10-minute experience.

3 EVALUATION & RESULTS

This study was conducted with the approval of the local ethics review board of the University of Tokyo.

User Experience with Interactive Body Editing. We surveyed 35 participants (6 female, 29 male; median age: 39.1; SD: 10.4) using the User Experience Questionnaire (UEQ) [3] at a local exhibition, where the environment was not strictly controlled. MBH scored *Excellent* in Attractiveness, Stimulation, and Novelty but received lower ratings for Perspicuity, Dependability, and Efficiency (Fig. 3). Participants appreciated the novelty of real-time body editing but found the system cumbersome at times (e.g., difficulties perceiving depth or unintentionally disconnecting wires). Some obstacles were intentionally introduced, such as reversed hand mappings (the user's right hand moving the avatar's left hand).

Impact on Body Perception and Acceptance. In a separate international exhibition, also under uncontrolled conditions, 59 participants (10 female, 49 male; median age: 32.1; SD: 11.9) completed a custom 7-point Likert questionnaire to assess changes in their perceptions of body identity, body-related expression, and impressions of body-editing technologies (Tab. 1). The results (Wilcoxon signed-rank test) showed a significant shift toward greater acceptance of body modification and fluid body identity, although no change was observed in satisfaction with one's innate body (Fig. 4). Participants' spontaneous comments revealed surprise at how quickly they adapted to altered control schemes, suggesting the experience may enhance short-term body plasticity.

Table 1: The questionnaire items.

Item	Statement
Q1	I am horrified to operate (even temporarily) a body that is different (in terms of manipulation, structure, and appearance) from my natural body as my body.
Q2	Just as one expresses one's identity by wearing the clothes and accessories one wants, I can express my identity by becoming the body I want to be.
Q3	I can personally accept technologies that allow me to manipulate a body different from my own (such as human augmentation technology, artificial limbs, prosthetics, or avatar technology).
Q4	The shape of my body is not likely to change, except for growth with age.
Q5	Only my innate body is truly my body.
Q6	My innate body is my identity.
Q7	I am satisfied with my body in daily physical activities.
Q8	In the future I may operate a body that is different from my innate body on a daily basis.
Q9	I think it would be great if everyone could edit their body at will.

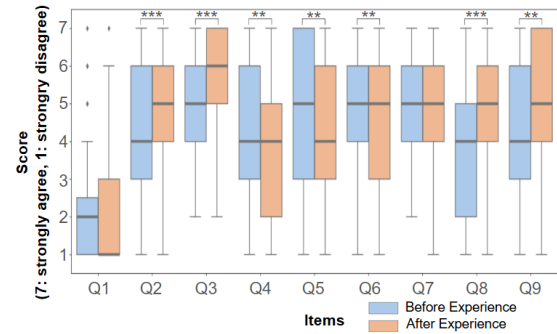


Figure 4: Results of the Questionnaire: **p < 0.01, ***p < 0.001

4 CONCLUSION & FUTURE WORKS

We demonstrated MBH as a novel body-editing interface, highlighting its experiential appeal and the resulting changes in user attitudes toward body modification. Further quantitative evaluation remains essential, particularly in exploring conditions where users (1) retain their original body, (2) add an extra limb, or (3) control a non-editable avatar. Another key direction would be to examine whether users can adapt to repeated transformations of their body schema over time. Insights from such research could enhance VR design and expand the applications of the "Editable Body" concept.

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